DATA 607 · PROJECT 3 · DEVELOPER SURVEY

Which data science skills are most valued?

How organizational differentiation shows up in pay and job satisfaction

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OUR DEFINITION

Value means organizational differentiation

A skill is valuable when it lets a person or team create a competitive advantage, not just execute routine work. We operationalize that across three dimensions.

Strategic impact vs. commodity work

Moves people from replaceable execution toward high-leverage work, such as end-to-end architecture.

Problem-solving complexity

Tackles unstructured, ambiguous problems that off-the-shelf automated AI tools cannot resolve.

Measurable ROI

Directly moves enterprise KPIs through better workflows, new revenue or insights competitors cannot copy.

DATA AND PIPELINE

Six steps from survey file to charts

Stack Overflow Developer Survey · 1,036 rows · 40 columns · analyzed in R with tidyverse and ggplot2

A

Ingest

Load the packages and read the CSV from GitHub with UTF-8 encoding.

B

Clean

Remove invalid IDs, non-positive pay and extreme outliers.

C

Parse skills

separate_rows() splits semicolon lists into one skill per row.

D

Pay premiums

Median and mean pay by role and skill set.

E

Satisfaction

Cross-reference job satisfaction with pay for each role.

F

Visualize

Publication-ready charts for the report.

CLEANING

From 1,036 raw rows to 168 analyzed respondents

1,036
Raw survey rows

1,033
Valid, unique ResponseId

191
US working professionals with pay between $20K and $1M

168
Classified as differentiating or execution roles

We keep US respondents only because CompTotal has no currency column, so pay from other countries cannot be compared.

OUTLIERS

We report medians, not means

All cleaned records (750)

Mean

$565,999

Median

$105,000

Low tool count (0–3 tools)

Mean

$804,553

Median

$90,000

MongoDB users

Mean

$772,091

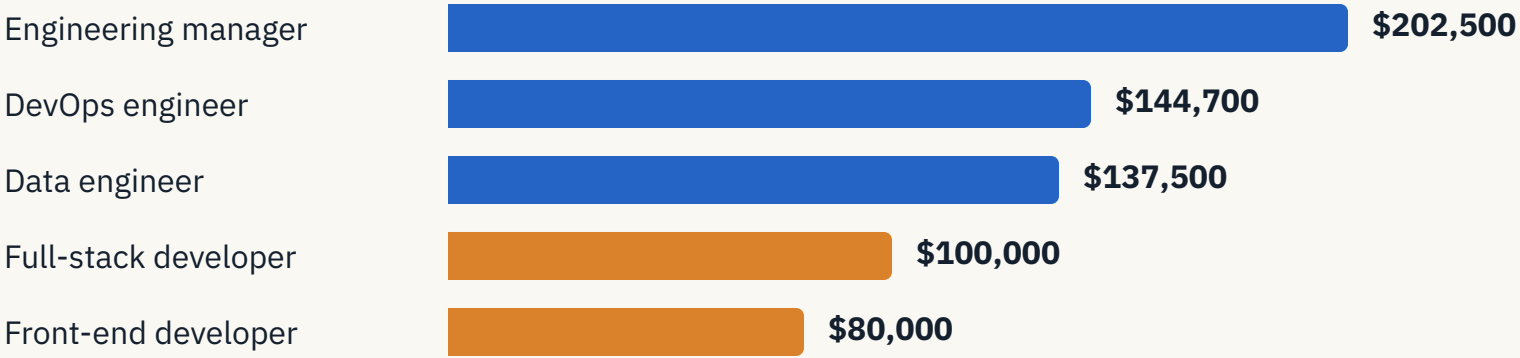
Median

$119,000

A few very large reported values pull the averages up, so the median is the better measure of typical pay.

FINDINGS · PAY

Specialized roles earn a premium over generalists

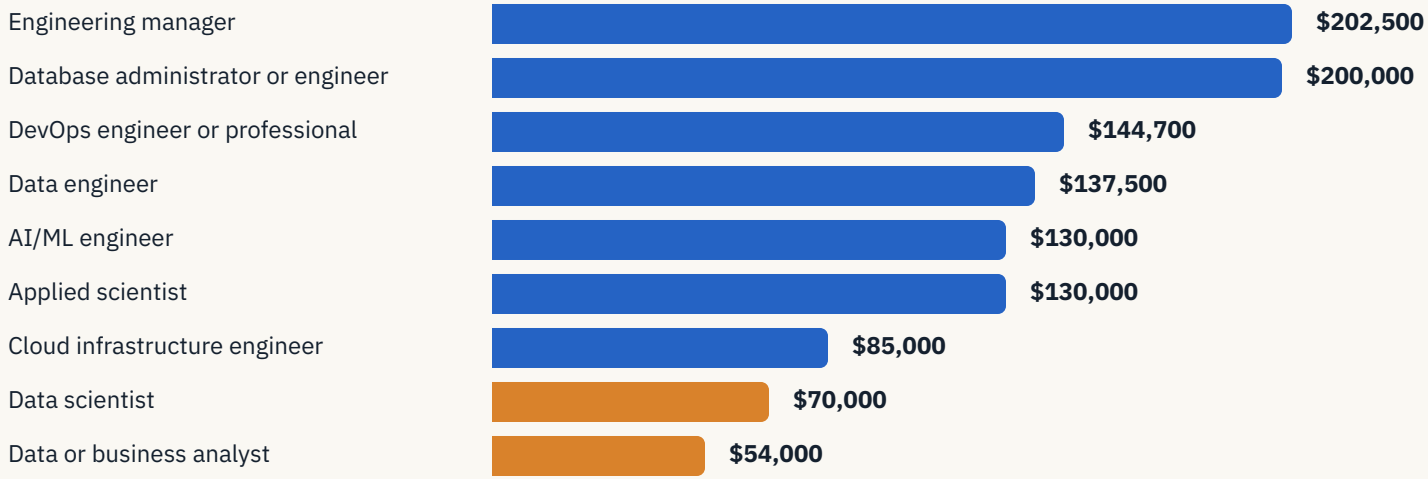


Blue: differentiating roles · **Orange**: generalist developer roles · median total compensation

DevOps and data engineers earn about 38% to 45% more than the $100,000 full-stack median.

FINDINGS · DATA ROLES

Roles that build systems out-earn roles that analyze data



Blue: manage, build or maintain systems · **Orange**: analysis-focused roles · median total compensation

FINDINGS · TECHNOLOGIES

Niche tools are paid more than the most-used ones

Languages

Language	Median pay
Go	$138,000
Ruby	$135,000
Python	$113,000
SQL	$112,500

SQL (460 respondents) and Python (417) are among the most-used languages.

Databases (25+ respondents each)

Database	Median pay
BigQuery	$170,000
Snowflake	$150,000
Supabase	$144,400
Cosmos DB	$140,000
PostgreSQL, MySQL, SQL Server, SQLite	$109,000 to $117,000
MariaDB	$92,500

Popularity is not the same as pay: cloud data platforms out-earn the general-purpose databases most respondents use.

FINDINGS · SATISFACTION

Satisfaction and pay are only loosely related

Engineering managers

Highest pay

7.65

average satisfaction

$202,500

median pay

Solutions architects

Highest satisfaction

7.70

average satisfaction

$120,000

median pay

Front-end developers

Lowest satisfaction

6.34

average satisfaction

$80,000

median pay

Satisfaction stays within 6.3 to 7.7 across roles, while median pay ranges from $80,000 to $202,500.

FINDINGS · SKILL GROUPS

Differentiating roles earn more

Differentiating (n=43)

$183,000

median annual pay

8.00

average job satisfaction (n=31)

Execution (n=125)

$138,000

median annual pay

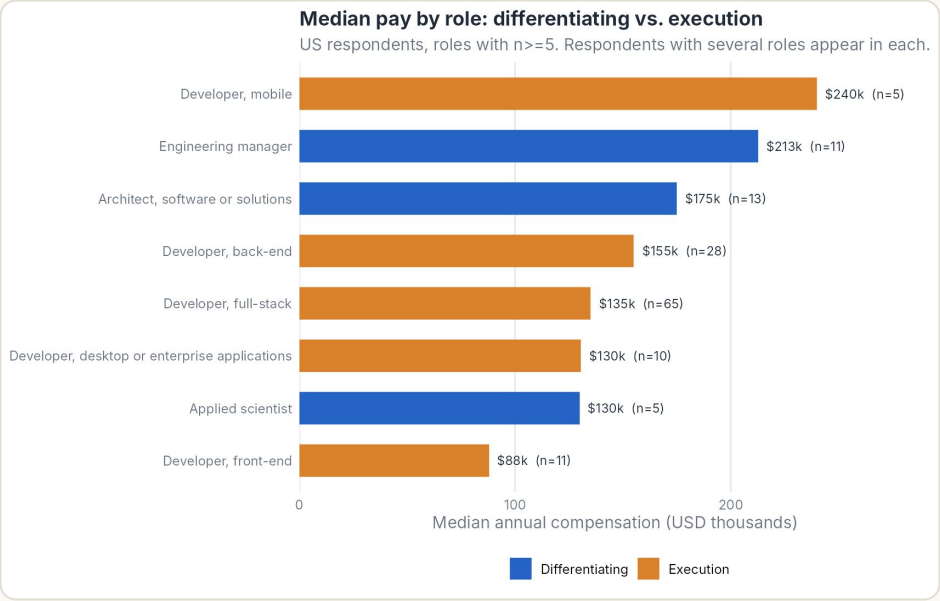
7.28

average job satisfaction (n=118)

A $45,000 gap in median pay, about 32.6% higher for differentiating roles.

GRAPH 1

Median pay by role



Managers and architects lead

Engineering managers ($213,000) and architects ($175,000) out-earn back-end ($155,000) and full-stack ($135,000) developers.

Not every differentiating role wins

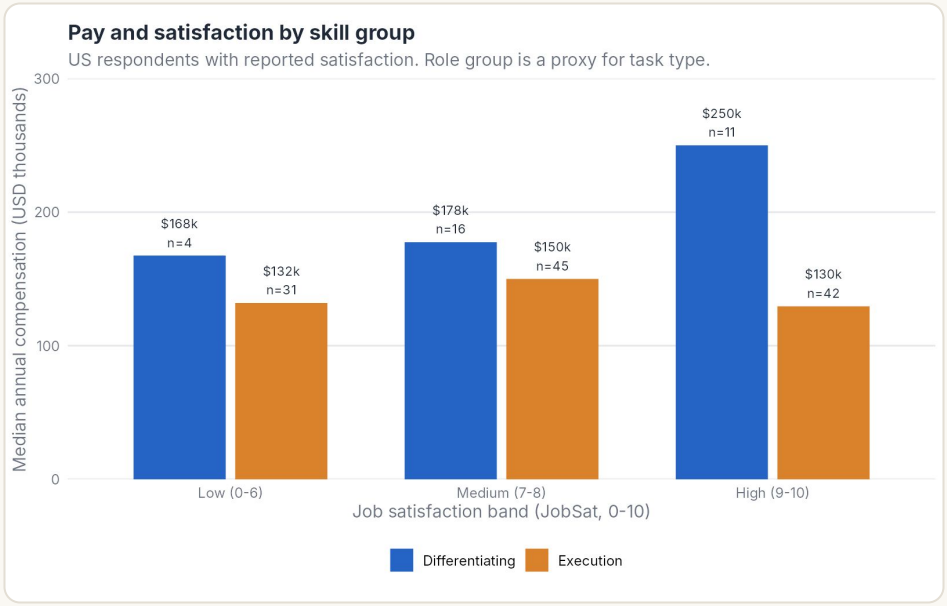
Applied scientists ($130,000) earn less than back-end developers, so pay varies by role, not only by group.

Read small samples with care

Mobile developers reach $240,000, but on only five respondents.

GRAPH 2

Pay and satisfaction by skill group



Higher pay in every band

Differentiating vs. execution: $168,000 vs. $132,000 (low), $178,000 vs. $150,000 (medium), $250,000 vs. $130,000 (high).

Widest gap at high satisfaction

The most satisfied differentiating respondents earn almost twice the execution median.

Read with care

Only 4 differentiating respondents are in the low band, and role group is a proxy for task type.

CONCLUSION

What the data suggests

Differentiation tracks pay

Median pay is $183,000 for differentiating roles and $138,000 for execution roles, about 32.6% higher.

Specialized beats popular

SQL and Python are widely used but earn less at the median than Go, Ruby, BigQuery or Snowflake.

Satisfaction follows, loosely

Differentiating respondents report higher satisfaction (8.00 vs. 7.28), but role-level patterns are mixed.

Limitations: the final sample is small (168 respondents, 43 differentiating), and role group is a proxy for task type. The results are consistent with our argument, not proof of it.